

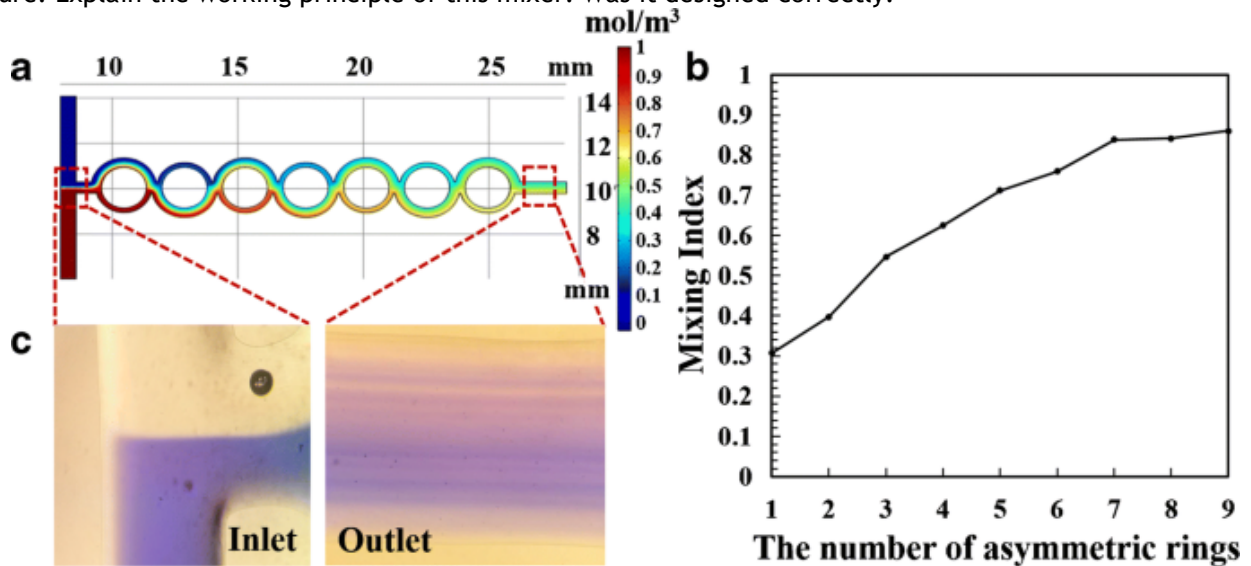
Politecnico di Milano
Biochip A.Y. 2019/2020 - M. Carminati
 On-Line Exam of 27.01.2021

Test duration: 2 hours - No support material admitted.

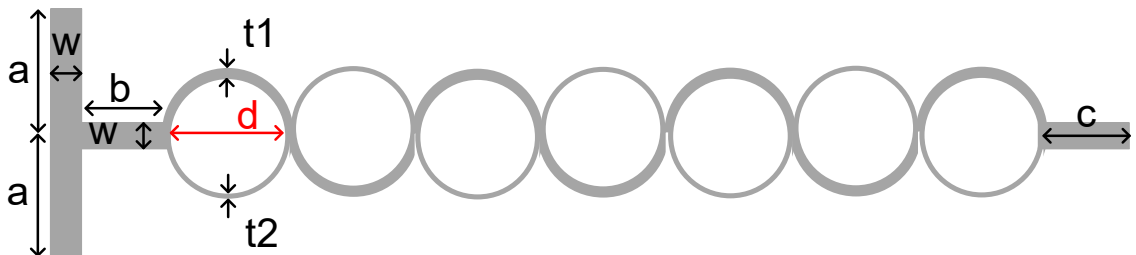
All the answers must be motivated. Beyond the final numerical results, all the expressions must be provided.

As usual the exam is composed of two parts: question 1-5 focus on the first system, while questions 6-12 on the second one. The link to upload the pdf is:

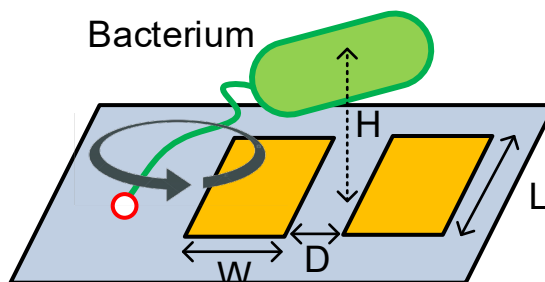
1. Consider the following microfluidic device implementing a mixer with two inlets (blue and red), as shown in the picture. Explain the working principle of this mixer. Was it designed correctly?



2. Compare this mixer with other types of mixers, in particular in terms of compactness, power consumption and mixing efficiency.
3. Draw the equivalent electrical model (and calculate the parameters) of the fluidic device considering the following dimensions: $a = 3\text{mm}$, $b = c = 1\text{mm}$, diameter $d = 2\text{mm}$, $w = 500\mu\text{m}$, $t_1 = 200\mu\text{m}$, $t_2 = 100\mu\text{m}$ and channel height $h = 100\mu\text{m}$ everywhere.



4. How does the model change if the liquid temperature changes?
5. If the pressure at both inlets is 2atm and the output is 1atm, what is the fluid speed in section c?
6. Now consider the following system to measure, by means of impedance, the **rotation of a single bacterium whose flagellum is tethered to the substrate** near a couple of coplanar sensing electrodes ($W = 4\mu\text{m}$, $L = 10\mu\text{m}$), at $H = 5\mu\text{m}$. Choose the optimal size of the distance D between the electrodes.



7. Assuming gold electrodes in PBS and a parasitic capacitance of 1pF between the electrodes, draw the impedance model between the electrodes and calculate the values of as many parameters of the model as possible.

8. Draw the scheme of the readout circuit and size the preamplifier considering a max. input current of $1\mu\text{A}$, taking the opamp ADA4817 (see an excerpt from the datasheet).

Parameter	Test Conditions/Comments	Min	Typ	Max	Unit
DYNAMIC PERFORMANCE					
-3 dB Bandwidth	$V_{\text{OUT}} = 0.1\text{ V p-p}$		1050		MHz
	$V_{\text{OUT}} = 2\text{ V p-p}$		200		MHz
	$V_{\text{OUT}} = 0.1\text{ V p-p}, G = 2$		390		MHz
Gain Bandwidth Product	$V_{\text{OUT}} = 0.1\text{ V p-p}$		≥ 410		MHz
Full Power Bandwidth	$V_{\text{IN}} = 3.3\text{ V p-p}, G = 2$		60		MHz
0.1 dB Flatness	$V_{\text{OUT}} = 2\text{ V p-p}, R_{\text{L}} = 100\ \Omega, G = 2$		60		MHz
Slew Rate	$V_{\text{OUT}} = 4\text{ V step}$		870		V/ μs
Settling Time to 0.1%	$V_{\text{OUT}} = 2\text{ V step}, G = 2$		9		ns
NOISE/HARMONIC PERFORMANCE					
Harmonic Distortion					
$f = 1\text{ MHz}$	$V_{\text{OUT}} = 2\text{ V p-p}, R_{\text{L}} = 1\text{ k}\Omega$				
			-113		dBc
			-117		dBc
$f = 10\text{ MHz}$	$V_{\text{OUT}} = 2\text{ V p-p}, R_{\text{L}} = 1\text{ k}\Omega$				
			-90		dBc
			-94		dBc
$f = 50\text{ MHz}$	$V_{\text{OUT}} = 2\text{ V p-p}, R_{\text{L}} = 1\text{ k}\Omega$				
			-64		dBc
			-66		dBc
Input Voltage Noise	$f = 100\text{ kHz}$		4		nV/ $\sqrt{\text{Hz}}$
Input Current Noise	$f = 100\text{ kHz}$		2.5		fA/ $\sqrt{\text{Hz}}$
DC PERFORMANCE					
Input Offset Voltage			0.4	2	mV

9. Compute and plot (power spectral density) the input-referred current noise of the circuit sized above.
10. Assuming a rotation speed of the bacterium of 10Hz, properly size the lock-in amplifier used to extract the impedance value.
11. Propose a solution to detect the rotation direction (clockwise or counter-clockwise) in addition to the rotation speed.
12. Design a new complete setup (electrodes + electronics) to implement a **differential** measurement to reject impedance drifts.